

Bhanuprakash Bhat

AWS Certified AI/ML Engineer | Data Science Enthusiast | Data Engineer | Agentic RAG

✉ bhanuprakashbhat@gmail.com ☎ 0432208480 📍 Sydney, Australia 🌐 LinkedIn 🐙 Github Profile

🔗 Portfolio (My works)

PROFILE

Data Scientist and Master's candidate at the University of Sydney with strong experience designing, deploying, and maintaining production-grade machine learning systems. Proven ability to architect end-to-end AI solutions, including a serverless Google Cloud Platform (GCP) pipeline that automated 100% of enterprise email triage, delivering an estimated ~\$10.6k annual productivity savings per employee.

Specialise in Retrieval-Augmented Generation (RAG) systems for complex information retrieval, leveraging vector databases, hybrid search (semantic + keyword), and LLM orchestration frameworks. Experienced in developing and deploying real-time ML pipelines using Python, FastAPI, Docker, and RESTful APIs, with a strong focus on scalability, reliability, and low-latency inference.

SKILLS

Programming: Python, R, Java, SQL, PostgreSQL, MySQL, Next Js

Data Analysis & Visualization:: Power BI (DAX, Power Query), Excel (Pivot Tables, Power Pivot), SQL (Joins, Subqueries, Window Functions), Python libraries (Seaborn, Matplotlib)

Machine Learning & AI Frameworks:: Scikit-learn, TensorFlow, PyTorch, Keras, OpenCV, XGBoost, LightGBM, LangChain, LangGraph, Huggingface, RAG solutions, Vector database

Software & Platforms:: Jupyter Notebooks, Google Colab, Git, GitHub, VS Code, MATLAB Simulink

PROFESSIONAL EXPERIENCE

Assessment and AI Transition Assistant

05/2025 – 12/2025 | Sydney, Australia

The University of Sydney

- Supported the University of Sydney's transition to AI-enabled learning and assessment systems by working across curriculum platforms, assessment data, and policy-aligned workflows.
- Collaborated with Educational Designers and Unit Coordinators to validate, standardise, and govern assessment metadata within the Sydney Curriculum Management System, ensuring data integrity, schema consistency, and policy compliance.
- Conducted assessment data quality audits, identifying structural inconsistencies, rule violations, and dependency issues across unit outlines.
- Contributed to change management and responsible AI adoption, supporting evaluation of emerging educational technologies and platform capability uplift.
- Assisted in documentation, stakeholder communication, and training activities to ensure smooth adoption of new AI-enabled assessment workflows.

Data Analyst

10/2024 – 03/2025 | Sydney, Australia

Clinic to Cloud

- Supported the migration of patient and clinical operational data from on-premise systems to a secure cloud-based data platform, ensuring data integrity, access control, and healthcare data governance compliance.
- Designed and implemented ETL pipelines to extract, clean, transform, and load patient demographics, visit histories, and clinical interaction data for analytics use cases.
- Structured and optimised datasets for analytics-ready schemas, improving data accessibility for reporting and downstream analysis
- Developed and delivered interactive dashboards for clinicians and operational staff to analyse patient visit trends, longitudinal health indicators, and clinic performance metrics.

Data Analyst Intern

- Designed and deployed dynamic Power BI dashboards to visualize critical sensor data (pressure, temperature, and density), enabling engineers and managers to track real-time trends and optimize system performance.
- Processed and cleaned large datasets using Pandas and NumPy, ensuring high data integrity before feeding into visualization pipelines and statistical analysis.
- Conducted exploratory data analysis (EDA) using Matplotlib and Seaborn to uncover hidden insights in system behavior, supporting early detection of anomalies in aerospace components.

PROJECTS

Serverless Autonomous AI Agent for Operational Triage on Google Cloud Platform

- **Architected a Serverless Pipeline:** Designed and deployed a containerized Python application on **Google Cloud Run (Jobs)**, utilizing **Docker** for cross-platform reproducibility and eliminating idle compute costs through scheduled execution.
- **AI-Driven Process Automation:** Integrated **OpenAI (GPT-4o)** to perform semantic analysis on real-time email streams (IMAP), successfully classifying urgency and routing actionable alerts to **Slack** via Webhooks with near-instant latency.
- **Business Impact & ROI:** Automated 100% of the manual email triage process, projecting a reduction in context-switching costs equivalent to **~\$10,600 in reclaimed annual productivity per employee..**

Autonomous RAG Compliance Agent: Australian Innovation Auditor (LangChain & LangGraph)

- Engineered an autonomous agent to audit R&D Tax, EMDG, and ESIC claims, reducing preliminary legal assessment costs by **~95%** (projected savings of **\$50,000+ annually** in external consulting fees).
- Built a custom **LangGraph** workflow utilizing **Hybrid Search** (Vector + Keyword) and ChromaDB to anchor every decision in official legislation, effectively eliminating hallucinations on critical financial caps.
- Developed a specialized "Grader" node to validate retrieval relevance, successfully flagging **100% of tested "Excluded Activities"** (e.g., Market Research & Internal Software) that typically trigger high-penalty regulatory audits.

Electricity Sector Data Integration & Spatial Analysis using DuckDB

- Built an end-to-end data engineering pipeline to integrate national electricity, emissions, and ABS regional datasets.
- Performed data cleaning, augmentation, and geocoding to validate facility coordinates and handle missing or malformed geographic data.
- Designed and implemented a spatial database schema in DuckDB with normalized staging tables and a denormalized analytical mart supporting geometry types.

EDUCATION

Master of Technology in Data Science and Artificial Intelligence

07/2024 – 06/2026 | Sydney, Australia

The University of Sydney

Bachelor of Technology in Aerospace Engineering

08/2020 – 06/2024 | Bengaluru, India

M S Ramaiah University of Applied Sciences

COURSES

Database Management Systems, Machine Learning Specialization , Principal of Data Science, Statistical Modelling and Probability, Data Mining, Data Engineering, Data Structures and Algorithms, Data Analysis, Object Oriented Programming, Project Management, Professional Practice in IT,

PUBLICATIONS

Three-Dimensional Computational Study over a Hyper-Ballistic missile body at Hypersonic Mach numbers
ISSW-34, Daegu, South Korea

14/07/2023